Part Two Concept Reference

Central concepts and notation for Chapters 1–12

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This reference gives one agreed meaning for each concept the twelve chapters of Part Two use. It exists so that a reader who meets a term in Chapter 3 can find the chapter and section that explains it, and so that the chapters need not define a term again each time they use it. The concept tables gather related terms into broad groups and give each term's meaning or essential distinction; the closing table records the notation the chapters share. It is written for lookup rather than for reading through. Chapters 1–4 are available on the website; references to Chapters 5–12 point forward to chapters that are not yet available.

A row is read from left to right. The preferred name comes first, with the common alternatives a reader will meet elsewhere beside it. The meaning cell states the essential distinction rather than a full explanation. The final cell of a concept table, **Main explanation**, is a destination: the chapter and section where the term is principally explained, with a first use elsewhere noted where one exists. In the notation table the **Main use** column carries that destination and a further column records alternative symbols and the cautions attached to them. Terms that name the same concept in context share an entry, related concepts with different meanings have separate entries, and a symbol whose preferred form carries a qualifier takes a row of its own that names its other uses. This reference records the vocabulary rather than teaching it: first-use explanations, worked examples and the reasons behind each distinction stay in the chapters themselves.

Latent Variables and the Common-Factor Model

Table 1
Latent variables and the common-factor model.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent-variable modeling	Latent variable (common factor, η_k)	An unobserved random variable introduced by a model and given a distribution over units. It is not a parameter, not a score, and not made observable by being named.	Chapter 1, Sec. 1.1
Latent-variable modeling	Construct	A substantive idea in the research question. Its meaning comes from theory, item content, response processes, population and intended use, none of which a covariance matrix supplies.	Chapter 1, Sec. 1.1
Latent-variable modeling	Indicator (item, observed response Y_j)	An observed variable proposed to bear on a factor; an item response is the value recorded for one unit.	Chapter 1, Sec. 1.1
Latent-variable modeling	Domain sampling (content or behavioral domain coverage)	Defensible coverage of the domain a construct names. It supports a broad construct interpretation and is usually systematic rather than a literal probability sample; it is a different question from sampling units from a population.	Chapter 1, Sec. 1.1
Latent-variable modeling	Reflective direction (formative alternative)	The modelling choice that writes each indicator as a function of the factors plus a residual. It states a direction of representation, not a causal finding, and it fails for constructs whose indicators produce them.	Chapter 1, Sec. 1.1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent-variable modeling	Classical true score ($X_{ir} = T_i + E_{ir}$)	The expected score of a person over repetitions of a specified procedure. It is a property of a person and a procedure, so a stable irrelevant influence stays inside it; it equals a common factor only under added assumptions.	Chapter 1, Sec. 1.2
Common-factor model	Common-factor model (linear factor model)	$\mathbf{y}_i = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i$ with the moment assumptions on $\boldsymbol{\eta}_i$ and $\boldsymbol{\epsilon}_i$. Its predictors are not columns of the dataset, which is what separates it from a regression.	Chapter 1, Sec. 2.1–2.2
Common-factor model	Loading (pattern coefficient, λ_{jk})	A fixed unknown parameter relating indicator j to factor k on a chosen factor metric. It is not a person's score and not a direct measure of how much attribute a person has.	Chapter 1, Sec. 2.1
Common-factor model	Residual (unique part, ϵ_j) and residual covariance θ_{jk}	The part of an indicator the declared factors do not represent, which may hold random fluctuation, item-specific content, method or occasion effects, an omitted factor, or misspecification. A residual covariance is a pair-specific relation outside the common part.	Chapter 1, Sec. 2.3
Common-factor model	Factor covariance matrix ($\boldsymbol{\Phi}$)	The $m \times m$ variance-covariance matrix of the factors, symmetric and positive semidefinite. It is total latent covariance, distinct from a disturbance covariance in a later structural model.	Chapter 1, Sec. 2.2
Common-factor model	Local independence (conditional independence)	Indicators being independent given the factors. It is stronger than zero conditional residual covariance, which is stronger than the zero unconditional covariance a diagonal $\boldsymbol{\Theta}$ supplies; none of the three says the observed items are unrelated.	Chapter 1, Sec. 2.3
Specification	Simple structure and cross-loading	A loading pattern in which each indicator relates to one factor, the remaining entries being fixed at zero; a cross-loading is a nonzero entry outside that pattern. Each zero is a restriction, not a drawing convenience. As a rotation criterion the same pattern favours relatively few substantial relations per item and identifiable indicator groups; it is not a criterion of fit to the observed covariance matrix, and rotation does not improve F_{ML}.	Chapter 1, Sec. 3.1; as a rotation criterion Chapter 2, Sec. 5.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Specification	Fixed, free and constrained parameter (parameter restriction)	Three distinct statuses an entry of $\boldsymbol{\Lambda}$, $\boldsymbol{\Phi}$ or $\boldsymbol{\Theta}$ can hold: fixed at a declared value, freely estimated, or tied to another parameter. A path diagram does not distinguish them.	Chapter 1, Sec. 3.2
Factor scale	Scale setting (marker variable, fixed factor)	Supplying a metric to a factor that has none, either by fixing one loading per factor or by fixing the factor variance. Neither convention is superior, and the implied covariance among indicators is the same under both.	Chapter 1, Sec. 4.1
Factor scale	Sign indeterminacy	Reversing a factor's direction and every one of its loadings leaves every product, and therefore the implied covariance, unchanged. A negative sign is a convention, and a sign flip between equivalent solutions is not a change in results.	Chapter 1, Sec. 4.2
Factor scale	Location convention (latent intercept $\boldsymbol{\alpha}$)	Fixing the origin of the factors, usually by setting the single-group factor means at zero, because observed means cannot determine unrestricted intercepts and unrestricted factor means together. $\boldsymbol{\alpha}$ is a latent intercept and coincides with $E(\boldsymbol{\eta}_i)$ only without structural paths.	Chapter 1, Sec. 4.2
Factor scale	Observed-unit change	Re-expressing recorded indicators in different units, which changes loadings, residual variances and implied covariances coherently and predictably. Nothing is indeterminate; this is not a latent-coordinate convention.	Chapter 1, Sec. 4.3
Factor scale	Rotational freedom (coordinate change, $\mathbf{Q}$)	The freedom, with two or more factors, to replace the axes while corresponding changes to $\boldsymbol{\Lambda}$ and $\boldsymbol{\Phi}$ leave the observed covariance unchanged. It resolves a representation choice and establishes nothing substantive.	Chapter 1, Sec. 4.1 and 7
Covariance structure	Model-implied covariance $\boldsymbol{\Sigma}(\boldsymbol{\vartheta})$ (covariance structure analysis)	$\boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}$: the covariance the model implies, split into a common and a residual part. It needs finite second moments and the stated moment restrictions, not normality.	Chapter 1, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Covariance structure	Pattern matrix and structure matrix ($\boldsymbol{\Lambda\Phi}$)	$\boldsymbol{\Lambda}$ holds the coefficients of the indicator equations; $\boldsymbol{\Lambda\Phi}$ holds indicator-factor covariances. They coincide under orthogonal unit-variance factors and must be read side by side in an oblique solution.	Chapter 1, Sec. 5; oblique reading Chapter 2, Sec. 6.1
Covariance structure	Communality (h_j^2)	The common part of an indicator's variance, $\boldsymbol{\lambda}_j^{\mathsf{T}}\boldsymbol{\Phi}\boldsymbol{\lambda}_j$, under the declared model. The shortcut λ_j^2 holds only for one nonzero loading on a unit-variance factor; correlated factors add a cross term.	Chapter 1, Sec. 5.1
Covariance structure	Uniqueness (residual variance; Heywood case)	$\theta_{jj} = 1 - h_j^2$ for a standardized indicator: variance the declared common factors do not represent, which is not the same as random error. A negative estimate is an improper, or Heywood, solution.	Chapter 1, Sec. 5.1
Covariance structure	Moment structure ($\boldsymbol{\mu}(\boldsymbol{\vartheta})$)	The modeled object is the mean and covariance pair; $\boldsymbol{\mu} = \boldsymbol{\nu} + \boldsymbol{\Lambda\alpha}$. Analysing covariances alone sets the mean structure aside, which is a choice rather than an absence.	Chapter 1, Sec. 5.2
Covariance structure	Distinct moment and degrees of freedom ($u = p(p+1)/2$, $df = u - t$)	The number of distinct entries a covariance matrix supplies, and its difference from the free-parameter count. More respondents estimate the same moments more precisely; $df \geq 0$ is necessary for, and never proof of, identification.	Chapter 1, Sec. 5.3
Covariance structure	Tetrad restriction	The vanishing of $\sigma_{jk}\sigma_{lm} - \sigma_{jm}\sigma_{lk}$, which one common factor with uncorrelated residuals forces for every four indicators. A nonzero tetrad refutes exact reproduction by that model, whatever its loadings.	Chapter 1, Sec. 6.3
Covariance structure	SS loading (column sum of squared loadings)	A summary of one factor column across indicators, read after rotation as a column statistic of the oriented solution. It is not in general an eigenvalue of the analysed correlation matrix, the matrix being decomposed must always be named, and with correlated factors it gives no unique additive partition of common variance.	Chapter 1, Sec. 6.1; post-rotation reading Chapter 2, Sec. 3.2
Neighboring objects	Factor-score estimate ($\hat{\eta}_{ik}$)	A prediction of an unobserved factor value from responses, a fitted model and a scoring rule. Different rules give different numbers from the same fit, and no rule makes it an error-free measurement.	Chapter 1, Sec. 6.5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Neighboring objects	Principal component and prespecified composite	Weighted composites of observed variables, one chosen to capture observed variance and one declared as a rule such as a sum or mean. Neither decomposes covariance into common and residual parts, so neither answers the common-factor model's question.	Chapter 1, Sec. 6.5
Analytic orientation	Exploratory, confirmatory and hybrid orientation	Names for how much is declared before the focal analysis, not for unrelated models. Both use the same covariance decomposition; the restrictions and the history of the decisions separate them.	Chapter 1, Sec. 1.4 and 6.6
Analytic orientation	Analytic status (development versus confirmation evidence)	The record of which body of evidence informed each decision. A specification chosen on one body of evidence receives a genuinely new test only from evidence not used to choose it; analyses move from confirmatory to exploratory status and never back.	Chapter 1, Sec. 6.6–6.7

Exploratory Factor Analysis and Model Development

Table 2
Exploratory factor analysis and model development.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Exploratory analysis	Exploratory factor analysis (EFA)	A common-factor analysis that leaves most loading restrictions open while seeking a comparatively low-dimensional representation of shared covariation. It is not assumption-free: the indicator set, population, association matrix, residual restriction, dimensional range and rotation criterion all shape what it can reveal.	Chapter 2, Sec. 1
Exploratory analysis	Development sample (development record)	The sample whose inspection produces a model. A model developed after inspecting these observations is exploratory with respect to them whatever fitting function is used, so the same records cannot confirm it.	Chapter 2, Sec. 1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Extraction	Squared multiple correlation (SMC)	For indicator j, the R^2 from regressing it on all other indicators, computed as $1 - 1/(\mathbf{R}^{-1})_{jj}$. It is a regression summary of the observed matrix used as a communality start, not a known common variance.	Chapter 2, Sec. 2.3
Extraction	Reduced correlation matrix (SMC-reduced matrix)	The correlation matrix with communality estimates replacing the unit diagonal. It need not be positive semidefinite, and its eigenvalues are on a different scale from the full matrix's.	Chapter 2, Sec. 2.3
Extraction	Principal-axis factoring (PAF)	Iterative extraction that decomposes a reduced matrix, forms loadings from its leading positive roots and updates the diagonal from row sums of squared loadings. The diagonal start, nonpositive-root handling and stopping rule are implementation choices.	Chapter 2, Sec. 2.3
Extraction	Minimum-residual extraction (minres)	Extraction minimizing the sum of squared off-diagonal correlation residuals, giving each displayed discrepancy equal weight. It is a common-factor method but not another name for PAF.	Chapter 2, Sec. 2.4
Retention	Kaiser–Guttman rule (eigenvalue-greater-than-one)	Retaining roots above one in a unit-diagonal correlation analysis. Sampling variation alone produces roots above one, and applying the threshold to a reduced matrix changes the comparison.	Chapter 2, Sec. 3.3
Retention	Scree plot	An ordered plot of eigenvalues read for where large decreases give way to a flatter sequence. It supplies no sampling reference, and full- and reduced-matrix plots need not agree.	Chapter 2, Sec. 3.3
Retention	Parallel analysis (permutation parallel analysis)	Comparing each observed eigenvalue with the same-rank eigenvalues of datasets built to have no common factors but the same sample size and marginal distributions, and retaining the leading consecutive run that clears its reference. The eigenvalue basis, the null generator, the reference statistic and the replication count and seed all belong to the specification.	Chapter 2, Sec. 3.4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Retention	Empirical Kaiser criterion (EKC)	A sample-adjusted repair of Kaiser–Guttman using the Marchenko–Pastur bound $(1 + \sqrt{p/n})^2$ for the first root, correcting later roots for variance already accounted for and flooring the reference at one.	Chapter 2, Sec. 3.5
Retention	Sequential model tests	Fitting successive factor counts and stopping at the first non-significant exact-fit test. The sequence is trustworthy only while still under-fitting, because the χ^2 reference fails once the count exceeds the true number.	Chapter 2, Sec. 3.5
Retention	Combination rule	Using two criteria with different references together, and treating their disagreement as evidence that the dimensions are hard to detect rather than as a tie to be broken by vote.	Chapter 2, Sec. 3.5
Counting	Effective EFA parameter count (t_m)	$pm + p - m(m-1)/2$ for an unrestricted common-factor covariance model at regular points, with $df_m = p(p+1)/2 - t_m$. The subtraction removes rotational redundancy; an oblique rotation adds no parameters because it re-expresses the same fitted covariance.	Chapter 2, Sec. 4.1
Counting	Ledermann bound	The largest factor count the covariance moments can generically support, $m \leq (2p + 1 - \sqrt{8p+1})/2$. It is a counting limit, not a recommendation, a positive-root count or a proof of identification.	Chapter 2, Sec. 4.1
Rotation	Inverse-paired transformation	The general nonsingular change of factor coordinates $\mathbf{\Lambda T}^{-1}$ with $\mathbf{T\Phi T}^{\mathsf{T}}$, which leaves the fitted common covariance unchanged. Multiplying loadings while leaving $\mathbf{\Phi}$ alone is not an equivalent oblique rotation.	Chapter 2, Sec. 5.1
Rotation	Varimax	An orthogonal criterion maximizing the dispersion of squared loadings within columns. A clean varimax pattern does not establish that the substantive attributes are unrelated.	Chapter 2, Sec. 5.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Rotation	Direct oblimin (quartimin at $\gamma = 0$)	An oblique criterion discouraging large squared coefficients in several columns of one row while permitting factor correlations. The name alone is not a specification: γ, normalization, tolerance, iteration limit, starts and seed all matter.	Chapter 2, Sec. 5.2
Rotation	Kaiser normalization	Dividing each loading row by its communality-based length before optimizing and restoring the scale afterward, which reweights indicators in the criterion. It is not the same operation as standardizing the observed variables.	Chapter 2, Sec. 5.2
Reading an oblique solution	Pattern coefficient (pattern matrix $\mathbf{P}$)	The coefficient of a factor in the simultaneous decomposition of a standardized indicator, read conditional on the other factors in the same representation.	Chapter 2, Sec. 6.1
Reading an oblique solution	Structure correlation (structure matrix $\mathbf{C} = \mathbf{P}\boldsymbol{\Phi}$)	The zero-order indicator–factor correlation, which can be appreciable where the pattern coefficient is near zero because part of the relation travels through the factor correlation. Pattern and structure coincide only in an orthogonal solution.	Chapter 2, Sec. 6.1 (defined as $\boldsymbol{\Lambda}\boldsymbol{\Phi}$ in Chapter 1, Sec. 5)
Reading an oblique solution	Oblique communality as a quadratic form	$h_j^2 = \mathbf{p}_j^{\mathsf{T}}\boldsymbol{\Phi}\mathbf{p}_j$, whose factor-correlation cross-products are part of the common variance. A row sum of squared pattern coefficients omits them and is a different quantity.	Chapter 2, Sec. 6.2
Residual evidence	Correlation residual matrix ($\mathbf{E}$) and off-diagonal RMS	Observed minus reproduced correlations, summarized by $\sqrt{2\sum_{j<k} e_{jk}^2/[p(p-1)]}$, which excludes variances. The largest absolute residual and the location of discrepancy carry information an average conceals, and this summary is not automatically any software statistic named RMSR or SRMR.	Chapter 2, Sec. 7.1
Residual evidence	Uniqueness boundary (bound-constrained uniqueness)	A fitted uniqueness at the implementation's lower bound rather than in the interior. It can signal too many factors, a redundant pair, sampling fluctuation or misspecification, and is different evidence from an interior estimate.	Chapter 2, Sec. 4.3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Preliminary summaries	KMO (Kaiser–Meyer–Olkin measure of sampling adequacy)	The ratio of summed squared zero-order correlations to that sum plus the summed squared partial correlations. It describes the input; it selects no factor count and is no minimum-sample-size calculation.	Chapter 2, Sec. 7.2
Preliminary summaries	Bartlett's test of sphericity	A normal-theory test of $H_0 : \mathbf{R}_{\text{population}} = \mathbf{I}$ on $p(p-1)/2$ df. Rejection says the indicators are not mutually uncorrelated, not that a low-dimensional factor model fits.	Chapter 2, Sec. 7.2
Analytic sensitivity	Tucker's congruence coefficient	The cosine between two loading-column vectors, compared after column order and sign are aligned. It compares coefficient directions, not participant scores, and near-one congruence can hide an important difference for one item.	Chapter 2, Sec. 7.3
Analytic sensitivity	Analytic sensitivity comparison	Changing one declared analytic choice at a time on the same records and comparing aligned solutions. It is same-sample evidence about analytic robustness, never an independent replication or a sampling interval.	Chapter 2, Sec. 7.3
Analytic status	Frozen candidate specification	The exact item order, free and fixed relations, scaling, covariance and mean settings, estimation family and named rivals recorded before any confirmation evidence is inspected. This separates a planned confirmation from a comparison developed after inspection, whose interpretation must acknowledge that selection.	Chapter 2, Sec. 8
Analytic status	Planned rival model	A restriction or broader alternative named before the confirmation outcome, distinguished from a post-hoc change, which changes analytic status and requires new evidence.	Chapter 2, Sec. 8

Confirmatory Specification, Identification, and Estimation

Table 3

Confirmatory specification, identification, and estimation.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Confirmatory specification	Confirmatory factor analysis (restricted factor model; CFA)	A common-factor model whose loading, residual, and factor-covariance entries are restricted before estimation, giving testable implications beyond the number of factors. The label describes restrictions and analytic history, not a favorable conclusion or a function name.	Chapter 3, Sec. 1
Confirmatory specification	Exclusion restriction (fixed zero loading; absent arrow)	A loading entry set to zero as a substantive claim that the indicator has no direct relation to that factor. It does not state that the indicator and factor are uncorrelated, because correlated factors still induce a nonzero structure coefficient.	Chapter 3, Sec. 1
Parameter status	Free, fixed, equality-constrained, bounded, and defined parameter	Estimated under the other restrictions; assigned a constant; required to share a value; restricted to a range; or computed from fitted parameters without adding freedom. A diagram shows which relations exist, not which status each has.	Chapter 3, Sec. 3
Parameter status	Residual covariance (correlated residual; correlated uniqueness)	A freely estimated off-diagonal entry of $\boldsymbol{\Theta}$, occupying two symmetric cells and adding one free parameter. It changes one indicator pair's implied covariance, unlike a cross-loading, which changes a whole row and column.	Chapter 3, Sec. 3
Latent metric	Marker scaling (reference indicator; unit loading identification)	Fixing one loading per factor at one, so the factor is expressed in that indicator's units and the factor variance is estimated. It does not set the factor variance to one and does not make the marker error-free.	Chapter 3, Sec. 4
Latent metric	Fixed-factor scaling (unit variance identification; `std.lv = TRUE`)	Fixing each factor variance at one and estimating every target loading, so a factor unit is one model standard deviation in a single-group model. It does not standardize the observed indicators.	Chapter 3, Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent metric	Factor-scale reparameterization ($\mathbf{D}$ transformation)	The diagonal rescaling $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{D}^{-1}$, $\mathbf{\Phi}^* = \mathbf{D}\mathbf{\Phi}\mathbf{D}^\mathsf{T}$, which leaves the implied covariance and $\mathbf{\Theta}$ unchanged. Equivalence fails if a constraint is not transformed with the metric or a factor variance is on a boundary.	Chapter 3, Sec. 4
Observed units	Observed-unit rescaling ($\mathbf{\Delta}$ rescaling)	Replacing $\mathbf{y}$ by $\mathbf{b} + \mathbf{\Delta}\mathbf{y}$, which multiplies an indicator's intercept and loading by its factor, its variance and residual variance by that factor squared, and its covariance with an unchanged indicator by the factor. It changes the record, not the latent coordinates, and leaves fit unchanged under a consistently transformed model.	Chapter 3, Sec. 4
Observed units	Correlation reparameterization (standardized-indicator model)	Writing $\mathbf{R} = \mathbf{D}_y^{-1/2}\mathbf{\Sigma}\mathbf{D}_y^{-1/2}$ with rescaled loading and residual matrices. It standardizes the indicators, not the factors, and its moment count differs from the covariance form because the unit diagonal is fixed.	Chapter 3, Sec. 4
Coefficient metrics	Latent-standardized and fully standardized loading (`std.lv`; `std.all`)	$\lambda\sqrt{\phi_{aa}}$ expresses a factor change in standard-deviation units in the indicator's raw unit; $\lambda\sqrt{\phi_{aa}}/\sqrt{\sigma_{jj}}$ standardizes both. A raw standard error is not the standardized coefficient's standard error, because the transformation uses estimated quantities.	Chapter 3, Sec. 4
Moment accounting	Observed moments, free parameters, and model degrees of freedom (u, t, $df = u - t$)	The distinct sample moments a declared representation supplies, the free coordinates the model estimates, and their difference. Necessary bookkeeping for one stated representation; neither identification nor quality, and unique covariance entries are not observations.	Chapter 3, Sec. 6
Moment accounting	Unrestricted-EFA dimension	The count of distinct covariance-model axes in an unrestricted m-factor solution, $pm + p - m(m-1)/2$ at regular full-rank points, which subtracts the continuous rotational freedom that changes the display without changing the reproduced covariance.	Chapter 3, Sec. 6

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Identification	Identification (global, local, and generic; empirical or weak identification)	Whether the moments determine the parameters: uniquely over the admissible space, in a neighborhood, or except at exceptional values. Weak identification describes a recoverable model whose parameters are poorly distinguished, showing as unstable estimates and very large standard errors rather than an error message.	Chapter 3, Sec. 6
Identification	Two- and three-indicator sufficient conditions (connectedness)	A factor with at least three exclusive indicators is identified under a fixed scale and diagonal residuals; a factor with two exclusive indicators needs a nonzero covariance with another identified factor. Both are sufficient and neither is necessary, and the second depends on an estimated quantity.	Chapter 3, Sec. 6
Identification	Moment Jacobian rank check	Full column rank of $\partial \mathbf{s}/\partial \boldsymbol{\vartheta}^{\mathsf{T}}$ at a regular fitted point, a local diagnostic that supports local identification. It does not prove global uniqueness or uniform stability, and it is a check on the fitted solution rather than a structural proof.	Chapter 3, Sec. 6
Model relations	Nested, nonnested, and equivalent models (observational equivalence)	Nesting requires that every covariance matrix one model generates belongs to the other's set, shown by the restrictions producing the inclusion. Equal degrees of freedom imply neither nesting nor equivalence, and observational equivalence limits what covariance data can distinguish.	Chapter 3, Sec. 6
Estimation	Normal-theory ML discrepancy (F_{ML})	$\log \lvert\boldsymbol{\Sigma}(\boldsymbol{\vartheta})\rvert + \operatorname{tr}\{\mathbf{S}\boldsymbol{\Sigma}(\boldsymbol{\vartheta})^{-1}\} - \log \lvert\mathbf{S}\rvert - p$, minimized over the admissible parameter space and normalized to zero at $\boldsymbol{\Sigma} = \mathbf{S}$. The covariance equation needs no normality; the conventional standard errors and test do.	Chapter 3, Sec. 7 (first used Chapter 2, Sec. 2.4)
Estimation	Covariance divisor convention ($\mathbf{S}_N$ against $\mathbf{S}_{N-1}$; normal against Wishart)	$\mathbf{S}_N = \{(N-1)/N\}\mathbf{S}_{N-1}$ is the normal-likelihood convention; the Wishart convention uses an $N-1$ covariance and multiplier. A supplied covariance matrix communicates neither the sample size nor the divisor, so the convention must be disclosed and never applied twice.	Chapter 3, Sec. 7

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Mean structure	Indicator intercepts and the latent location convention	Free indicator intercepts with latent intercepts fixed at zero add p moments and p parameters, leaving the covariance restrictions and degrees of freedom unchanged. Freeing both at once identifies no latent origin, because $\boldsymbol{\alpha} + \mathbf{c}$ with $\boldsymbol{\nu} - \boldsymbol{\Lambda}\mathbf{c}$ reproduces the same means.	Chapter 3, Sec. 5
Solution quality	Convergence, admissibility, and the Heywood case	Convergence is an optimizer's stopping statement; admissibility requires positive residual variances and positive definite factor and implied covariance matrices. A negative or zero residual variance is a symptom of the whole decomposition and is neither a cosmetic warning nor something to replace by zero in a report.	Chapter 3, Sec. 8
Inference	Wald test of a linear restriction ($\mathbf{R}\boldsymbol{\vartheta} = \mathbf{r}$)	A χ^2_q test computed from the estimates and their covariance matrix, requiring the covariance between estimates and not only their variances. It tests the stated restriction conditional on the rest of the model; it is not a global-fit test and need not equal a refitted likelihood-ratio statistic.	Chapter 3, Sec. 8
Transition record	Evaluation transition record (handoff record)	The auditable record of evidence identity, model, parameter conventions, estimation settings, numerical checks, parameter uncertainty, global/local evaluation and its limits, and reproduction metadata. It preserves what was fixed, inspected, or revised so a later comparison cannot silently change the analysis.	Chapter 3, Sec. 8
Observed system	Parceling (item aggregation)	Fitting $\mathbf{z} = \mathbf{A}\mathbf{y}$ with $\mathrm{Var}(\mathbf{z}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\mathsf{T}$. Reducing dimension lets different item-level patterns give the same parcel covariance, so a well-fitting parcel model does not establish the item-level restrictions.	Chapter 3, Appendix

Evaluation, Comparison, Reliability, and Extensions

Table 4

Evaluation, comparison, reliability, and extensions.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Global fit	Model chi-square (exact-fit test; likelihood-ratio model test)	$T_M = N\widehat{F}_{\mathrm{ML}}$ referred to $\chi^2_{df_M}$, testing whether some admissible parameter vector reproduces the population moment structure exactly. Nonrejection is not acceptance of the model, and rejection at large N is evidence against exact fit rather than a measure of its magnitude.	Chapter 3, Sec. 8; Chapter 4, Sec. 2
Global fit	RMSEA (root mean square error of approximation) with its 90% interval	Estimated discrepancy per degree of freedom, $\sqrt{\max\{(T_M - df_M)/(N\,df_M), 0\}}$, whose behavior also depends on model size and degrees of freedom, with an interval obtained by inverting a noncentral chi-square reference. A zero estimate is a truncation, not zero residuals; the convention (N or $N-1$) must be named.	Chapter 3, Sec. 8; Chapter 4, Sec. 2
Global fit	Incremental fit index (CFI; TLI or NNFI) with its independence baseline	Improvement of the model relative to a stated reference rather than relative to exact fit. The usual baseline frees the observed variances and fixes all covariances to zero, so a high value depends on how poor that baseline is; TLI is not bounded by one.	Chapter 3, Sec. 8; Chapter 4, Sec. 2
Global fit	SRMR (standardized root mean square residual)	Root mean square of observed-SD-standardized covariance residuals over one triangle including the diagonal. Not a mean absolute residual; a small value can hide a few large cells among many near-zero ones.	Chapter 3, Sec. 8; Chapter 4, Sec. 3
Local diagnosis	Covariance residual (raw; observed-SD-standardized; correlation residual)	Observed minus implied covariance, either in the product of the two indicators' units, divided by the product of observed SDs, or as a difference of separately formed correlations. None is a residual divided by its sampling standard error, so none is automatically a z statistic.	Chapter 3, Sec. 8; Chapter 4, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Local diagnosis	Modification index (score statistic; Lagrange multiplier) and expected parameter change (EPC; standardized EPC)	A local one-parameter approximation to the improvement from releasing one fixed restriction, with the approximate value the released parameter would take. Read with the residual pattern, the parameter's meaning, and the status of the search; not the exact LR change, and not an instruction.	Chapter 4, Sec. 6
Comparison	Nested difference test (χ^2 difference; likelihood-ratio difference test)	$\Delta T = T_R - T_U$ on $\Delta df = df_R - df_U$ for regularly nested, compatible ML fits on identical cases, outcomes, scale, mean structure, and likelihood convention. A smaller parameter count is not proof of nesting, and failure to reject is not equivalence.	Chapter 4, Sec. 4
Comparison	Information criteria for covariance structures (AIC; BIC) eligibility	$-2\ell + 2q$ and $-2\ell + q \log N$ with q the effective free-parameter count. They rank eligible candidates fitted to the same observations and outcomes under one likelihood convention, including nonnested ones; neither is a test or an adequacy index.	Chapter 4, Sec. 4
Comparison	Score, Wald, and likelihood-ratio views of one restriction	Three statistics for one null: slope and information at the restricted fit; distance of the unrestricted estimate from the restriction relative to its uncertainty; difference of optimized likelihoods. Asymptotic equivalence is not finite-sample equality.	Chapter 4, Sec. 4
Comparison	Equivalent models (covariance-equivalent representation)	Distinct specifications with the same set of admissible implied covariance matrices, so that each is nested in the other: every covariance matrix one can reproduce, the other can also reproduce, as an exact parameter mapping shows. On any data they therefore give the same fitted covariance matrix, statistic, df, and fit indices. Identical fit in one sample is a consequence, not a test, and no covariance data can choose between them; only theory and evidence beyond reproduced covariance can.	Chapter 4, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Extensions	Higher-order factor model (second-order factor; first-order disturbance)	A model in which one further latent variable accounts for the covariances among first-order factors, $\boldsymbol{\Phi}_l = \boldsymbol{\gamma}\boldsymbol{\gamma}^{\mathsf{T}} + \boldsymbol{\Psi}_h$. The first-order disturbances are distinct from indicator residuals. With diagonal first-order disturbances, the generic covariance-equality restriction count is $m(m-3)/2$ for $m \geq 3$: three factors add no equality degrees of freedom, but nonnegative disturbance variances still restrict the admissible correlations.	Chapter 4, Sec. 5
Extensions	Bifactor model (general plus orthogonal specific or group factors)	Every item loads directly on one general factor and one assigned specific factor, all factors mutually orthogonal. A specific factor is common variance among its items after the general factor is held constant, not an item residual, and the model is generally nonnested against a correlated-factor model of the same items.	Chapter 4, Sec. 5
Extensions	Schmid-Leiman transformation and the proportionality restriction	The algebraic rewriting of a higher-order model as an orthogonal general factor with loadings $\boldsymbol{\Lambda}_l\boldsymbol{\gamma}$ and specific factors with loadings $\boldsymbol{\Lambda}_l\boldsymbol{\Psi}_h^{1/2}$. Because both come from one loading matrix, each simple-structure item in a domain has the same general-to-specific loading ratio $\gamma_k/\psi_{h,k}^{1/2}$; a freely estimated bifactor model is not bound by that ratio.	Chapter 4, Sec. 5
Extensions	Method effect; testlet effect	Common variance among items sharing a format (reverse keying, response scale, informant) or a stimulus (passage, vignette, apparatus). Both are design-known groupings and the least controversial motivation for a specific factor, which generalizes a freed residual covariance to several items.	Chapter 4, Sec. 5
Reliability	Cronbach's alpha for a declared unit-weighted score, α_C	A coefficient computed from item variances and their total-score variance. Interpreting it as reliability requires additional measurement assumptions; it is not a universal lower bound when residual dependence is allowed.	Chapter 4, Sec. 8

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Reliability	Model-based omega for a declared score, $\omega(\mathbf{w})$ (composite reliability)	The proportion of a declared weighted composite's model-implied variance attributable to all modeled common factors. It retains cross-loadings, factor correlations, and modeled residual covariances, and is conditional on model, weights, and sample.	Chapter 4, Sec. 8
Reliability	Omega-hierarchical, $\omega_h(\mathbf{w})$	The share of a declared score's variance attributable specifically to a bifactor general factor. Its complement within total omega is modeled reliable specific-factor variance, not error, and a substantial value does not establish essential unidimensionality.	Chapter 4, Sec. 5
Reliability	Average variance extracted (AVE) and coefficient H as bounded neighbours	AVE summarizes average communality for a simple fully-standardized single-factor item set; $H = a/(1+a)$ with $a = \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{\Theta}^{-1}\boldsymbol{\lambda}$ is a maximum over optimally weighted composites. Neither is the reliability of a stated unit-weighted score, and neither applies unqualified to a cross-loading subscale.	Chapter 4, Appendix
Scores	Factor score (regression or conditional-mean score; Bartlett score) and factor-score determinacy	A model-dependent linear prediction of a latent variable from the indicators, whose weights use the whole fitted structure; determinacy is the model-based correlation between factor and prediction. A score is a prediction with nonzero conditional variance, not an observed latent value.	Chapter 4, Sec. 8
Analytic history	Pseudo-true (population-ML) parameter and population discrepancy (F_0; population RMSEA)	The optimum of a misspecified model fitted to the exact generating covariance, and the discrepancy remaining there. Displacement at this optimum is deterministic under two stated specifications, not sampling fluctuation and not finite-sample estimator bias.	Chapter 4, Sec. 7
Analytic history	Post-selection revision and independent revalidation	A change suggested by inspecting a body of evidence is exploratory with respect to that evidence, whatever function fits it; only records not used to suggest it can support it, and a retest that fails to support it does not prove the parameter is zero.	Chapter 4, Sec. 6

Observed Structural Systems

Table 5
Observed Structural Systems.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Structural systems	Structural equation	An equation representing an outcome as a function of its designated predictors and a disturbance, within a declared system. Its causal interpretation requires assumptions beyond covariance fit.	Chapter 5, Sec. 1
Structural systems	Endogenous variable	A variable explained by another variable in the structural system. This model role differs from econometric endogeneity, which concerns dependence between a regressor and its equation's disturbance.	Chapter 5, Sec. 1; Sec. 3
Structural systems	Exogenous variable	A variable whose distribution is supplied without modeling incoming structural paths in the declared system. The label alone does not establish independence from every unmeasured cause.	Chapter 5, Sec. 1
Structural systems	Structural disturbance (ζ)	Variation in a structural outcome not represented by its designated predictors. Covariances among disturbances and with exogenous variables are part of the model's assumptions.	Chapter 5, Sec. 1
Structural systems	Reduced form	The structural equations solved in terms of the exogenous inputs and disturbances, when the required inverse exists. Reduced-form coefficients combine direct and transmitted relationships.	Chapter 5, Sec. 1; Sec. 7
Structural systems	Direct, indirect and total effect	A direct coefficient belongs to one structural equation; an indirect contribution follows a directed route. In an acyclic linear system, the total sums the compatible directed-route products. Feedback requires the inverse-system response. A causal reading requires the declared causal assumptions.	Chapter 5, Sec. 2; Sec. 7
Structural systems	Saturated structural system	A structural parameterization that leaves the modeled variables' covariance unrestricted. Its coefficients can be estimated with uncertainty even though its global covariance test has zero degrees of freedom.	Chapter 5, Sec. 3; Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Structural systems	Recursive and nonrecursive system	Under the chapter's convention, a recursive linear system has an acyclic ordering, mutually uncorrelated disturbances, and disturbances uncorrelated with exogenous inputs. Feedback or relaxed disturbance assumptions require a separate identification analysis.	Chapter 5, Sec. 1; Sec. 7
Feedback	Feedback (reciprocal system)	A directed cycle in a simultaneous structural system. Its equilibrium interpretation differs from an observed lagged relationship across measurement occasions.	Chapter 5, Sec. 7
Feedback	Equilibrium stability	For the declared repeated linear adjustment, the condition $\rho(\mathbf{B}) < 1$ that makes departures decay. Existence of $(\mathbf{I} - \mathbf{B})^{-1}$ alone does not establish stability.	Chapter 5, Sec. 7
Reproducibility	Summary-statistic reconstruction	Analysis using the published moments, variable definitions, sample information and model restrictions actually available. Distinguish a numerical reproduction from a changed reanalysis or an independent empirical replication.	Chapter 5, Sec. 5; Sec. 8

Causal Models, Interventions and Identification

Table 6
Causal Models, Interventions and Identification.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Causal interpretation	Structural causal model (SCM)	Structural assignments and a distribution for their disturbances, with an intervention defined by replacing an assignment while retaining the other declared mechanisms.	Chapter 6, Sec. 1
Causal interpretation	Conditioning	Comparing or selecting observations at a recorded value of a variable. It generally changes the distribution of related variables and is different from intervening on the variable.	Chapter 6, Sec. 1
Causal interpretation	Intervention	Replacing a variable's structural assignment by a specified value or policy under the model's invariance assumptions. Setting a value removes that variable's incoming causal arrows.	Chapter 6, Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Causal interpretation	Collider	A variable at which two arrows on a path meet head to head. Conditioning on it or its descendant can open an otherwise blocked path.	Chapter 6, Sec. 2
Causal interpretation	d-separation	A graph criterion that blocks every path between two sets under a conditioning set and implies conditional independence under the graph's Markov assumptions. An open path permits dependence but does not guarantee it for every parameter value.	Chapter 6, Sec. 2
Identification	Instrumental variable	A source of regressor variation requiring relevance, exclusion from the outcome equation, and the stated independence or orthogonality to its disturbance. A first-stage association tests neither exclusion nor exogeneity.	Chapter 6, Sec. 6
Identification	Instrument relevance	Association with the endogenous regressor after the included controls are accounted for. Weak conditional association makes the identifying covariance ratio unstable.	Chapter 6, Sec. 6
Identification	Exclusion restriction	A restriction removing a direct route from an instrument to the outcome in the stated structural equation and control set. It needs substantive support distinct from first-stage strength.	Chapter 6, Sec. 6
Identification	Causal identification	Whether the stated causal target is uniquely determined by the observed distribution together with the declared causal assumptions; statistical parameter recovery within one chosen model does not establish this.	Chapter 6, Sec. 1; Sec. 6
Adjustment	Backdoor adjustment	Recovering an intervention distribution by averaging conditional outcome distributions over a stated target covariate distribution when the covariates satisfy the causal adjustment requirements, with consistency and positivity.	Chapter 6, Sec. 4
Counterfactuals	Counterfactual outcome	The outcome under a specified replacement of a structural assignment while retaining the unit's relevant disturbances; factual evidence determines what is known or uncertain about those disturbances.	Chapter 6, Sec. 5
Counterfactuals	Abduction, action and prediction	Updating disturbance information using factual evidence, replacing the specified assignment, and propagating that retained information through the altered system.	Chapter 6, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Sensitivity	Exclusion sensitivity	Re-estimating a target under explicitly fixed alternative direct instrument-outcome effects; conditional sampling intervals and the across-assumption sensitivity range answer different questions.	Chapter 6, Sec. 7

Joint Measurement and Structural Modeling

Table 7
Joint Measurement and Structural Modeling.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Full SEM	Joint measurement model	One model containing all relevant measurement blocks, with their factors allowed to covary before structural restrictions are imposed.	Chapter 7, Sec. 1
Full SEM	Two-step SEM workflow	Evaluation of a joint measurement specification followed by joint estimation of its measurement and structural parameters in the full SEM.	Chapter 7, Sec. 1
Full SEM	Structurally saturated model	A model whose structural component reproduces an unrestricted covariance among its latent variables; measurement restrictions may still leave the complete model with positive degrees of freedom.	Chapter 7, Sec. 2; Sec. 4
Exploratory measurement	ESEM	A structural equation model containing an exploratory factor block whose estimated loadings are oriented by rotation, potentially alongside confirmatory blocks and structural regressions.	Chapter 7, Sec. 5
Exploratory measurement	Target rotation	Rotation toward a declared loading pattern whose target entries guide orientation rather than necessarily fixing corresponding loadings to exact values.	Chapter 7, Sec. 5
Measurement representation	Causal indicator	An observed variable proposed to contribute to a latent variable whose disturbance represents additional causes. Identification requires more than assigning arrows to its predictors.	Chapter 7, Sec. 6
Measurement representation	Exact composite	A variable defined as a specified weighted combination of observed components, with no disturbance in that defining equation.	Chapter 7, Sec. 6

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Means and covariates	Mean structure	The model's specification of expected observed levels through intercepts, latent means, and predictor means.	Chapter 7, Sec. 7
Means and covariates	MIMIC model	A model combining observed predictors of a latent variable with multiple indicators of that latent variable.	Chapter 7, Sec. 7

Indirect and Conditional Relationships

Table 8

Indirect and Conditional Relationships.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Indirect relationships	Specific indirect effect	The product of coefficients along one directed indirect route between a stated source and outcome in the declared linear structural system.	Chapter 8, Sec. 1
Indirect relationships	Total indirect effect	The sum of the specified directed indirect-route contributions between the same source and outcome. Contributions with opposite signs can cancel.	Chapter 8, Sec. 2
Indirect relationships	Indirect-effect contrast	A difference or other declared comparison of indirect effects whose uncertainty retains the dependence between their estimated component paths.	Chapter 8, Sec. 2
Conditional relationships	Conditional slope	The rate of change in a modeled conditional mean with a predictor at stated values of the moderator and other conditioned variables. Its value depends on the declared scale and origin.	Chapter 8, Sec. 6
Conditional relationships	Latent interaction	A product relationship involving latent predictors represented through explicit measurement models. A latent outcome alone does not turn an observed predictor product into a latent interaction.	Chapter 8, Sec. 5
Conditional relationships	Product indicator	An observed product of indicators used to represent a latent product under specified measurement-error and centering assumptions. Its measurement error contains more than the product of the two original errors.	Chapter 8, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Conditional relationships	Conditional indirect effect	A specified indirect-effect function evaluated at stated moderator values, with its uncertainty propagated through the joint parameter estimates.	Chapter 8, Sec. 7
Conditional relationships	Index of moderated mediation	The coefficient describing change in a specified indirect effect per unit of a moderator when that indirect-effect function is linear in the moderator. Its formula depends on the structural model.	Chapter 8, Sec. 7

Measurement Comparability and Group Comparisons

Table 9

Measurement Comparability and Group Comparisons.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Comparability	Measurement invariance	Equality of the conditional response distribution across groups at the same latent values. Particular loading, intercept and residual restrictions express aspects of this requirement within a declared measurement model.	Chapter 9, Sec. 1; Sec. 3
Comparability	Configural model	A joint group model with the same designated factor-loading pattern and separately estimated group parameters under declared scale and location constraints. The shared pattern alone does not establish a common factor unit or origin.	Chapter 9, Sec. 2
Comparability	Metric invariance (loading invariance)	Equality of the designated factor loadings across groups, supporting a common factor unit through the stated anchors. It does not impose common indicator intercepts or residual structures.	Chapter 9, Sec. 3
Comparability	Scalar invariance (intercept invariance)	Equality of the designated indicator intercepts as well as loadings, supporting a common origin for the specified latent-mean comparison. The comparison group's latent means remain free when they are the target.	Chapter 9, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Comparability	Strict invariance	Equality of the measurement residual covariance structure in addition to the scalar restrictions. It is not generally required for a latent-mean comparison and does not alone guarantee equal reliability.	Chapter 9, Sec. 3
Comparability	Partial invariance	A declared model that retains specified cross-group equality restrictions while allowing selected measurement parameters to differ. Its interpretation depends on the remaining anchors and the evidence supporting them.	Chapter 9, Sec. 4
Identification and comparison	Anchor	A retained measurement restriction that supplies a common reference for the latent scale or origin. A numerically identifying anchor can still lack substantive comparability.	Chapter 9, Sec. 4
Identification and comparison	Anchor sensitivity	Variation in an interpreted factor comparison when the declared parameters linking its scale or origin across groups are changed.	Chapter 9, Sec. 4
Group targets	Latent-mean contrast	A difference or other declared comparison of latent means on a supported common scale and origin. Its standardized form additionally depends on the chosen reference variance.	Chapter 9, Sec. 5
Group targets	Multigroup SEM	Joint estimation of group-specific measurement and structural systems with explicit restrictions on which parameters are shared and which can differ.	Chapter 9, Sec. 2; Sec. 5
Group targets	Conditional group contrast	A comparison at specified covariate values under the fitted model. It can differ from the marginal group difference because the groups' covariate distributions differ.	Chapter 9, Sec. 5; Sec. 6
Item comparisons	Intercept differential item functioning (uniform DIF)	A group difference in an indicator's conditional expected response at the same latent values, represented by an intercept difference or a group-to-indicator direct path in the stated model. It is distinct from loading noninvariance.	Chapter 9, Sec. 6

Incomplete Data and Robust Inference

Table 10
Incomplete Data and Robust Inference.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Observation process	Missingness pattern	The combination of observed and missing variables in a record.	Chapter 10, Sec. 1
Observation process	MCAR	Missingness does not depend on the observed or unobserved values in the analysis.	Chapter 10, Sec. 2
Observation process	MAR	Conditional on the observed information included in the analysis, missingness does not depend on the unobserved values.	Chapter 10, Sec. 2
Observation process	MNAR	Missingness still depends on unobserved values after conditioning on the observed information in the analysis.	Chapter 10, Sec. 2
Observation process	Ignorability	Conditions under which inference for the data-model parameters can omit an explicit missingness model; for likelihood inference these include MAR and distinct parameters.	Chapter 10, Sec. 2
Observed-data estimation	Observed-data likelihood	A likelihood formed from the values actually observed, integrating over the missing values under the specified model.	Chapter 10, Sec. 3
Observed-data estimation	FIML	Joint maximum-likelihood estimation using each record's observed variables and the corresponding model-implied distribution.	Chapter 10, Sec. 3
Additional information	Auxiliary variable	An additional variable included to supply information about incomplete values or missingness without assigning it a substantive structural role.	Chapter 10, Sec. 6
Robust inference	Sandwich covariance	A parameter covariance estimate combining likelihood curvature with the empirical variability of individual score contributions.	Chapter 10, Sec. 5
Robust inference	Scaled difference test	A model-comparison test that adjusts the difference in compatible unscaled fit statistics using information from the fitted models.	Chapter 10, Sec. 5
Additional information	Sensitivity parameter	A quantity fixed at substantively considered values to examine how conclusions depend on an assumption the observed data do not identify.	Chapter 10, Sec. 6

Latent Growth Models

Table 11
Latent Growth Models.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Trajectory components	Latent intercept	A person-specific level of the modeled trajectory at the declared time origin. Its unit and location depend on the outcome or measurement scale.	Chapter 11, Sec. 2
Trajectory components	Latent slope	A person-specific change per unit of coded time in a linear growth model; its meaning changes when the time unit or growth basis changes.	Chapter 11, Sec. 2
Trajectory components	Mean slope	The population mean of the latent slope in an unconditional model; with predictors, the structural slope intercept describes the mean at the stated predictor reference.	Chapter 11, Sec. 2
Trajectory components	Slope variance	Between-person variance of the latent slope under the specified trajectory model, distinct from uncertainty in the estimated mean slope.	Chapter 11, Sec. 2
Time and shape	Intercept–slope covariance	Covariance between the latent level at a declared time origin and the latent slope; changing that origin can change its value or sign without changing the modeled trajectories.	Chapter 11, Sec. 4
Time and shape	Time centering	Replacing time by time minus a stated constant, changing the intercept meaning and its moments while preserving the same trajectory model.	Chapter 11, Sec. 4
Time and shape	Latent basis	A growth model that fixes two basis loadings to set the origin and unit and estimates the remaining loadings; people share the basis shape while their levels and amplitudes vary.	Chapter 11, Sec. 4
Time and shape	Piecewise growth	A trajectory represented by distinct functions over prespecified time segments, with explicit restrictions on the knot, continuity and the meaning of each slope.	Chapter 11, Sec. 4
Longitudinal measurement	Occasion disturbance	An occasion-specific latent deviation from the growth curve, distinguished from an indicator-specific measurement residual in a multiple-indicator growth model.	Chapter 11, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Longitudinal measurement	Longitudinal measurement comparability	Support for measurement restrictions that make the intended construct comparison meaningful across occasions, including suitable loading and intercept anchors for latent mean growth.	Chapter 11, Sec. 5

Study Planning and Power Analysis

Table 12
Study Planning and Power Analysis.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Study planning	Prospective power	Probability that a specified test rejects its null under a stated alternative, study design and analysis.	Chapter 12, Secs. 1 and 3
Study planning	Parameter power	Power for a named parameter restriction within a specified model.	Chapter 12, Secs. 1 and 3
Study planning	Global-fit power	Power for a stated hypothesis about the model's overall population discrepancy.	Chapter 12, Sec. 3
Study planning	Calibration scenario	A fully specified planning population informed by evidence and explicit choices.	Chapter 12, Sec. 2
Simulation	Monte Carlo error	Numerical uncertainty because performance is estimated from finitely many simulated studies.	Chapter 12, Sec. 5
Simulation	Operating characteristic	A feature of an analysis's repeated-study performance under a declared scenario.	Chapter 12, Sec. 5
Simulation	Coverage	Probability that the stated interval procedure contains the target value across repeated studies under the scenario.	Chapter 12, Sec. 5
Simulation	Simulation failure rate	Fraction of scheduled analysis attempts that fail the declared convergence, admissibility or usable-output criteria.	Chapter 12, Sec. 5
Study planning	Precision target	A declared requirement for the uncertainty of the study's estimate.	Chapter 12, Secs. 1 and 8

Notation for Chapters 1–12

Table 13
Notation for Chapters 1–12.

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Indices and counts	i, j, k; n, p, m	Unit, indicator and factor indices; numbers of units, indicators and factors	Chapter 1, Sec. 2.1	$p = 6$ and $m = 2$ throughout Chapter 1's running system; N writes the analysis sample size in the likelihood conventions of Chapter 3, Sec. 7 and Chapter 4, Sec. 2; a also serves as a factor index in Chapters 2–3
Observed	y_{ij}, $\mathbf{y}_i$, Y_j	One response, the $p \times 1$ response vector for unit i, and the indicator as a random variable	Chapter 1, Sec. 2.1–2.2	Uppercase only where the distinction from an observed value matters
Model	$\boldsymbol{\nu}$, ν_j	The $p \times 1$ vector of indicator intercepts	Chapter 1, Sec. 2.1–2.2	—
Model	$\boldsymbol{\Lambda}$, $\boldsymbol{\lambda}_j$, λ_{jk}	The $p \times m$ loading (pattern) matrix, its jth row as a column vector, and the entry relating indicator j to factor k	Chapter 1, Sec. 2.2, 3	Rows are indicators and columns are factors; $\boldsymbol{\Lambda}_{\text{simple}}$ for the restricted pattern; $\boldsymbol{\Lambda}_l$ is the first-order loading matrix of Chapter 4, Sec. 5; Part One's notation uses λ for the Poisson rate (Part One Chapters 2 and 12) and the penalty size (Part One Chapter 13), so the loading reading is stated at first use

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Model	$\boldsymbol{\eta}_i$, η_{ik}	The $m \times 1$ vector of common factors for unit i and one factor value	Chapter 1, Sec. 2.1	$\boldsymbol{\eta}_i^*$ for a rescaled, sign-reversed, shifted or rotated version; η is the GLM linear predictor in Part One's notation (Part One Chapter 11) and η^2 an ANOVA effect size (Part One Chapter 4), so the factor reading is stated at first use
Model	$\boldsymbol{\epsilon}_i$, ϵ_{ij}	The $p \times 1$ residual vector and one residual	Chapter 1, Sec. 2.1	Never called measurement error without an argument
Model	$\boldsymbol{\alpha}$	The $m \times 1$ latent intercept vector; the factor mean only where no structural paths exist	Chapter 1, Sec. 2.2	With $\mathbf{B}$ present, $E(\boldsymbol{\eta}_i) = (\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\alpha}$; no $\boldsymbol{\kappa}$ convention; Part One's notation uses α for the Type I error probability (Part One Chapter 3) and the elastic-net mixing parameter (Part One Chapter 13)
Model	$\boldsymbol{\Phi}$, ϕ_{jk}	The $m \times m$ factor covariance matrix and its entries; $\boldsymbol{\Phi}_{2\rho}$ for the correlated two-factor case	Chapter 1, Sec. 2.2, 6.4	Total latent covariance across the series, not a disturbance covariance $\boldsymbol{\Psi}$; $\boldsymbol{\Phi}_l$ is the first-order factor covariance matrix of Chapter 4, Sec. 5; Part One's notation uses Φ for the standard Normal distribution function (Part One Chapter 2)
Model	$\boldsymbol{\Theta}$, θ_{jk}	The $p \times p$ residual covariance matrix and its entries	Chapter 1, Sec. 2.2–2.3	$\mathrm{diag}(\theta_{11}, \ldots, \theta_{pp})$ for the diagonal working version; the parameter space is $\mathcal{P}$, never another Θ; Part One's notation uses θ, $\hat{\theta}$ for a generic estimand (Part One Chapter 3) and the negative-binomial dispersion (Part One Chapter 12)

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Implied moments	$\boldsymbol{\vartheta}$, $\boldsymbol{\Sigma}(\boldsymbol{\vartheta})$, $\boldsymbol{\mu}(\boldsymbol{\vartheta})$	The free-parameter vector and the implied covariance and mean it produces	Chapter 1, Sec. 5	$\boldsymbol{\Sigma}$ alone where the argument is clear; Chapter 3, Sec. 1 reads $\boldsymbol{\vartheta}$ as the complete free-parameter vector, distinct from $\boldsymbol{\Theta}$
Implied moments	h_j^2	The communality of indicator j under the declared model	Chapter 1, Sec. 5.1	$\theta_{jj} = 1 - h_j^2$ for a standardized indicator; written as the quadratic form $\mathbf{p}_j^{\mathsf{T}}\boldsymbol{\Phi}\mathbf{p}_j$ in Chapter 2, Sec. 6.2 and for a cross-loading indicator in Chapter 3, Sec. 8; u_j^2 writes the matching uniqueness in Chapter 2
Counting	u, t, df	Distinct observed moments, free parameters, and their difference	Chapter 1, Sec. 5.3	df upright; $u = 21$ for six indicators; reused in Chapter 3, Sec. 6; the subscripted u_j^2 of Chapter 2 is a uniqueness, not a moment count; an unsubscripted t here is the free-parameter count, distinct from the Part One t_{df} reference distribution (Part One Chapter 3)
Counting	u_{cov}, $u_{\text{mean+cov}}$	Moment counts without and with an unrestricted mean vector	Chapter 3, Sec. 5	Free indicator intercepts with latent intercepts fixed add equally to moments and parameters, leaving df unchanged
Counting	t_m	The effective parameter count of an unrestricted m-factor solution, $pm + p - m(m-1)/2$	Chapter 2, Sec. 4.1	The unrestricted-EFA dimension of Chapter 3, Sec. 6 is the same count; an unsubscripted t is the free-parameter count of Chapter 1, Sec. 5.3

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Coordinates	a, $\mathbf{c}$, $\mathbf{Q}$	A nonzero rescaling constant, an $m \times 1$ location shift, and an orthogonal coordinate matrix	Chapter 1, Sec. 4.1–4.2, 7	$\mathbf{T}$ for the general invertible transformation is Chapter 2's; $\mathbf{Q}$ is the orthogonal rotation matrix of Chapter 2, Sec. 5.2; a also indexes factors in p_{ja}, λ_{ja} and ϕ_{aa} (Chapters 2–3) and names $\boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{\Theta}^{-1}\boldsymbol{\lambda}$ in coefficient H (Chapter 4, Appendix), so the setting is stated at first use
Coordinates	$\mathbf{T}$	The general nonsingular change of factor coordinates, $\boldsymbol{\Lambda}\mathbf{T}^{-1}$ paired with $\mathbf{T}\boldsymbol{\Phi}\mathbf{T}^{\mathsf{T}}$	Chapter 2, Sec. 5.1	Reused for the rotational family in Chapter 3, Sec. 6; the orthogonal case keeps Chapter 1's $\mathbf{Q}$
Coordinates	γ, δ	The oblimin parameter and the geomin parameter of the rotation criteria	Chapter 2, Sec. 5.2	Plain letters; the bold $\boldsymbol{\gamma}$ and $\boldsymbol{\delta}$ of Chapter 4, Sec. 5 are second-order loadings and first-order disturbances
Classical test theory	X_{ir}, T_i, E_{ir}	Observed score on repetition r, true score, and error	Chapter 1, Sec. 1.2	$E_r(\cdot \mid i)$ averages over repetitions with the person fixed
Scores	$\widehat{\eta}_{ik}$	A factor-score estimate for unit i on factor k	Chapter 1, Sec. 6.5	The hat marks a prediction from a fitted model, never the factor; $\mathbf{A}_R$ and $\mathbf{A}_B$ of Chapter 4, Appendix supply the weights
Fixed arrays	$\mathbf{I}_p$, $\mathbf{1}_p$, diag$(\cdot)$	The $p \times p$ identity, the $p \times 1$ vector of ones, and the diagonal matrix formed from the listed entries	Chapter 1, Sec. 6.1–6.2	$\mathbf{1}_6$ writes the one-factor record's equal loadings as $.7\,\mathbf{1}_6$

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Sample records	$\mathbf{S}$, $\mathbf{R}$	The sample covariance matrix, divisor $n-1$, and the sample Pearson correlation matrix	Chapter 2	Reserved in Chapter 1, which has no sample and uses $\mathbf{S}$ only for its rescaling anchor; the structure matrix is written $\mathbf{\Lambda\Phi}$ and given no symbol; Part One's notation homes both as observed-variable matrices and, like Part Two, writes matrices and vectors bold with a superscript T for the transpose
Sample records	$\mathbf{D}_S$	The diagonal matrix of sample variances	Chapter 2	$\mathbf{D}_y$ is the observed-variance counterpart in the correlation reparameterization of Chapter 3, Sec. 4
Sample records	$\mathbf{V}$, $\mathbf{D}$, d_k	Eigenvectors, the eigenvalue matrix, and the kth eigenvalue of $\mathbf{R}$	Chapter 2	A spectral eigenvalue matrix; the $\mathbf{D}$ of Chapter 3, Sec. 4 is the diagonal factor-scale transformation and Part One's notation uses $\mathbf{D}$ for a diagonal SD matrix, $\widehat{\mathbf{V}}$ for estimated coefficient covariance (Part One Chapter 8) and d for Cohen's d (Part One Chapter 3), so the setting is named at first use
Sample records	$\mathbf{R}_c^{(t)}$	The reduced correlation matrix at iteration t	Chapter 2, Sec. 2.3	The superscript is an iteration index, not the free-parameter count t; the reduced matrix need not be positive semidefinite

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Extraction	$\mathbf{A}$, $\mathbf{A}_m$	The unrotated loading matrix and its rank-m PCA counterpart	Chapter 2	$\mathbf{A}$ is the parcel weight matrix in Chapter 3, Appendix and $\mathbf{A}_R$, $\mathbf{A}_B$ are factor-score weight matrices in Chapter 4, Appendix; each use is declared locally
Extraction	F_{minres}	The minimum-residual extraction discrepancy	Chapter 2, Sec. 2.4	A common-factor criterion, not another name for principal-axis factoring
Retention	SS_k	The sum of squared loadings in column k	Chapter 2, Sec. 3.2	Generally not a spectral eigenvalue, and distinct from the Part One ANOVA SS family
Retention	B, $Q_{.95}$	The number of permutation replications and the 95th-percentile null reference	Chapter 2, Sec. 3.4	The bold $\mathbf{B}$ of Chapter 4, Sec. 8 is the latent structural path matrix
Retention	EKC_j	The empirical Kaiser reference for root j	Chapter 2, Sec. 3.5	Built on the Marchenko–Pastur bound $(1 + \sqrt{p/n})^2$ and floored at one
Retention	T_m, df_m	The corrected exploratory model statistic for m factors and its degrees of freedom	Chapter 2, Sec. 3.5	Lowercase m marks the exploratory sequence; the uppercase T_M, df_M of Chapter 4, Sec. 2 are the confirmatory model statistic and its degrees of freedom
Rotation	$\mathbf{P}$, p_{ja}	The pattern matrix and the pattern coefficient of indicator j on factor a	Chapter 2, Sec. 6.1	$\mathbf{\Lambda}$ names the same object in the model equations of Chapter 1 and Chapter 3
Rotation	$\mathbf{C}$	The structure matrix $\mathbf{C} = \mathbf{P\Phi}$ of indicator–factor correlations	Chapter 2, Sec. 6.1	Defined as $\mathbf{\Lambda\Phi}$ in Chapter 1, Sec. 5, which gives it no symbol

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Residual evidence	$\mathbf{E}$, e_{jk}, $\mathrm{RMS}_{\mathrm{off}}$	The correlation residual matrix, its entries, and the off-diagonal summary $\sqrt{2\sum_{j<k} e_{jk}^2/[p(p-1)]}$	Chapter 2, Sec. 7.1	In Chapter 4, Sec. 3 e_{jk} is the raw covariance residual and $e_{jk}^{(s)}$, $e_{jk}^{(r)}$ the standardized and correlation scales, so the scale is always named; this summary is not automatically any software RMSR or SRMR
Residual evidence	$\hat{r}_{jk}$	A reproduced correlation	Chapter 2, Sec. 7.1	The hat marks a model-implied value, not a sample statistic
Preliminary summaries	a_{jk}	A partial correlation entering the KMO ratio	Chapter 2, Sec. 7.2	Distinct from the rescaling constant a of Chapter 1, Sec. 4.1 and from the factor index a
Preliminary summaries	T_B, df_B	Bartlett's sphericity statistic and its $p(p-1)/2$ degrees of freedom	Chapter 2, Sec. 7.2	In Chapter 4, Sec. 2 the same pair names the independence-baseline statistic of the incremental indices, a different null
Analytic sensitivity	$c(\mathbf{a}, \mathbf{b})$	Tucker's congruence coefficient between two loading columns $\mathbf{a}$ and $\mathbf{b}$	Chapter 2, Sec. 7.3	Compared after column order and sign are aligned; these are loading columns, not the shift vector $\mathbf{b}$ of Chapter 3, Sec. 4
Parameter space	$\mathcal{P}$	The admissible parameter space	Chapter 3, Sec. 7	Never written Θ, which is the residual covariance matrix
Latent metric	$\mathbf{D}$ (factor scale)	The diagonal factor-scale transformation, $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{D}^{-1}$ with $\mathbf{\Phi}^* = \mathbf{D}\mathbf{\Phi}\mathbf{D}^{\mathsf{T}}$	Chapter 3, Sec. 4	Carries its qualifier, because $\mathbf{D}$ is also the eigenvalue matrix of Chapter 2
Latent metric	$\lambda_{ja}^{(\mathrm{lv})}$, $\lambda_{ja}^{(\mathrm{all})}$	Latent-standardized and fully standardized loadings	Chapter 3, Sec. 4	$\lambda\sqrt{\phi_{aa}}$ and $\lambda\sqrt{\phi_{aa}}/\sqrt{\sigma_{jj}}$; a raw standard error is not the standardized coefficient's standard error

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Observed units	$\boldsymbol{\Delta}$, $\mathbf{b}$	The diagonal observed-unit conversion matrix and its shift vector, giving $\mathbf{b} + \boldsymbol{\Delta}\mathbf{y}$	Chapter 3, Sec. 4	Reserved for observed-unit rescaling across Part Two and named without algebra in Chapter 1; the moment Jacobian is written $\partial\mathbf{s}/\partial\boldsymbol{\vartheta}^{\mathsf{T}}$, never $\boldsymbol{\Delta}$. Chapter 3, Sec. 6.5 writes the moment Jacobian in that form, and the Chapter 4 local-rank check inherits it; Part Two keeps the two symbols separate
Observed units	$\mathbf{D}_y$	The diagonal matrix of observed variances used in the correlation reparameterization	Chapter 3, Sec. 4	$\mathbf{R} = \mathbf{D}_y^{-1/2}\boldsymbol{\Sigma}\mathbf{D}_y^{-1/2}$; $\mathbf{D}_S$ of Chapter 2 is the sample counterpart
Observed units	$\mathbf{R}$ (implied correlation)	The model-implied correlation matrix	Chapter 3, Sec. 4	Carries its qualifier: an unqualified $\mathbf{R}$ is the sample correlation matrix of Chapter 2
Identification	$\mathbf{s}(\boldsymbol{\vartheta})$, $\partial\mathbf{s}/\partial\boldsymbol{\vartheta}^{\mathsf{T}}$	The stacked implied moments and their Jacobian	Chapter 3, Sec. 6	Full column rank at a regular fitted point supports local identification only
Estimation	F_{ML}	The normal-theory ML discrepancy	Chapter 3, Sec. 7 (first used Chapter 2, Sec. 2.4)	$\widehat{F}_{\mathrm{ML}}$ at the fitted point enters $T_M = N\widehat{F}_{\mathrm{ML}}$ in Chapter 4, Sec. 2; rotation does not change it
Estimation	$\mathbf{S}_N$, $\mathbf{S}_{N-1}$	The sample covariance matrix under the two divisor conventions, $\mathbf{S}_N = \{(N-1)/N\}\mathbf{S}_{N-1}$	Chapter 3, Sec. 7	$\mathbf{S}_N$ is reused in Chapter 4, Sec. 2; the convention is disclosed and never applied twice
Inference	$\mathbf{R}$ (restriction), $\mathbf{r}$, q, W	The restriction rows, their values, their number, and the Wald statistic for $\mathbf{R}\boldsymbol{\vartheta} = \mathbf{r}$	Chapter 3, Sec. 8	Carries its qualifier; this q counts restrictions, while the q of Chapter 4, Sec. 4 is an effective free-parameter count

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Parcels	$\mathbf{A}$ (parcel), $\mathbf{z}$	The parcel weight matrix and the parcel score vector, with $\mathrm{Var}(\mathbf{z}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^{\mathsf{T}}$	Chapter 3, Appendix	Carries its qualifier beside the extraction and factor-score uses of $\mathbf{A}$
Global fit	T_M, df_M	The model test statistic $N\widehat{F}_{\mathrm{ML}}$ and the model degrees of freedom of the declared covariance model	Chapter 4, Sec. 2	Uppercase M; the lowercase T_m, df_m of Chapter 2, Sec. 3.5 belong to the exploratory sequence
Global fit	T_B (baseline), df_B	The independence-baseline statistic and its degrees of freedom in CFI and TLI	Chapter 4, Sec. 2	Carries its qualifier; the same pair is Bartlett's sphericity test in Chapter 2, Sec. 7.2
Local diagnosis	e_{jk}, $e_{jk}^{(s)}$, $e_{jk}^{(r)}$	Raw, observed-SD-standardized, and correlation covariance residuals	Chapter 4, Sec. 3	None is divided by a sampling standard error, so none is a z statistic; e_{jk} is the correlation residual in Chapter 2, Sec. 7.1, so the scale is named
Comparison	ΔT, Δdf	The nested difference statistic and its degrees of freedom	Chapter 4, Sec. 4	For regularly nested, compatible ML fits on identical cases, outcomes, scale, mean structure, and likelihood convention
Comparison	$\ell(\widehat{\boldsymbol{\vartheta}})$, q	The maximized log likelihood and the effective free-parameter count in $-2\ell + 2q$ and $-2\ell + q\log N$	Chapter 4, Sec. 4	This q is a parameter count; the q of Chapter 3, Sec. 8 counts restriction rows
Extensions	$\boldsymbol{\gamma}$	The $m \times 1$ second-order loading vector	Chapter 4, Sec. 5	Bold; the plain γ of Chapter 2, Sec. 5.2 is the oblimin parameter
Extensions	$\boldsymbol{\delta}$, $\boldsymbol{\Psi}_h$, $\psi_{h,k}$	First-order disturbances, their diagonal covariance matrix, and its kth diagonal entry, with $\boldsymbol{\Phi}_l = \boldsymbol{\gamma}\boldsymbol{\gamma}^{\mathsf{T}} + \boldsymbol{\Psi}_h$	Chapter 4, Sec. 5	Bold; the plain δ of Chapter 2, Sec. 5.2 is the geomin parameter. First-order disturbances are not indicator residuals
Extensions	$\boldsymbol{\Lambda}_l$, $\boldsymbol{\Phi}_l$	The first-order loading matrix and first-order factor covariance matrix of the higher-order form	Chapter 4, Sec. 5	The subscript l marks the first-order level

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Extensions	λ_{jg}, λ_{js}; G, S_{is}	Bifactor general and specific loadings for item j; the general factor and the specific factor for unit i in domain s	Chapter 4, Sec. 5	S_{is} is a specific factor, not an entry of the sample covariance matrix $\mathbf{S}$
Reliability and scores	$\mathbf{w}$, c_i	Declared score weights and the composite for unit i	Chapter 4, Sec. 8	The weights are declared before any coefficient is calculated
Reliability and scores	$\omega(\mathbf{w})$, $\omega_h(\mathbf{w})$	The model-based common-variance proportion for those weights and omega-hierarchical	Chapter 4, Secs. 8 and 5	Both are conditional on model, weights, and sample
Reliability and scores	H	Coefficient H under a restrictive one-factor model, $H = a/(1+a)$ with $a = \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{\Theta}^{-1}\boldsymbol{\lambda}$	Chapter 4, Appendix	Plain; Part One's notation uses the bold $\mathbf{H}$ for the regression hat matrix
Reliability and scores	$\mathbf{A}_R$, $\mathbf{A}_B$	Regression and Bartlett factor-score weight matrices	Chapter 4, Appendix	$\hat{\eta}_{ik}$ of Chapter 1, Sec. 6.5 is the resulting prediction
Population audit	$\boldsymbol{\Sigma}_0$, $\boldsymbol{\Sigma}^*$	The exact generating covariance and its pseudo-true fitted counterpart	Chapter 4, Sec. 7	Displacement at the pseudo-true optimum is deterministic, not sampling fluctuation
Population audit	F_0, RMSEA_0	The population discrepancy and the population RMSEA $\sqrt{F_0/df}$	Chapter 4, Sec. 7	Not the sample RMSEA formula
Structural extension	$\mathbf{B}$, $\boldsymbol{\Gamma}$, $\boldsymbol{\zeta}$, $\boldsymbol{\Psi}$	Latent structural paths, exogenous links, the structural disturbance, and its covariance	Chapter 4, Sec. 8	Bold $\mathbf{B}$; the plain B of Chapter 2, Sec. 3.4 is a replication count. The bare $\boldsymbol{\Psi}$ is the structural disturbance covariance and nothing else: the Schmid-Leiman equations of Chapter 4, Sec. 5.5 write the first-order disturbance covariance as $\boldsymbol{\Psi}_h$